A Distributed Real-Time Intrusion Detection System on a Jetson Orin Nano Super Edge Cluster using Kafka and PySpark

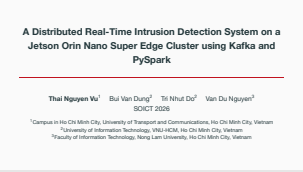
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SOICT 2026

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2026-10-20

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Good morning. I will show what a second edge board actually buys you when you split an ML intrusion detector across two Jetson Orin Nano Super devices, and what it costs in detection. Eight minutes, five results.

I. PROBLEM AND CONTRIBUTIONS

Edge intrusion detection: three open gaps

Detectors now *fit* on single-board hardware — but a monolithic one classifies **every** flow and saturates an 8 GB ARM board.

What the literature does not tell us

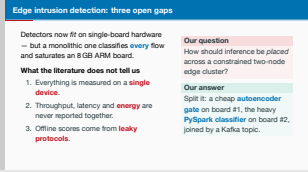
1. Everything is measured on a **single device**.
2. Throughput, latency and **energy** are never reported together.
3. Offline scores come from **leaky protocols**.

Our question

How should inference be *placed* across a constrained two-node edge cluster?

Our answer

Split it: a cheap **autoencoder gate** on board #1, the heavy **PySpark classifier** on board #2, joined by a Kafka topic.

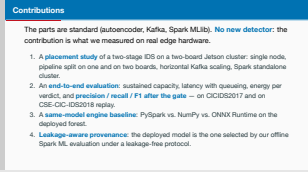


Three gaps: single-device evaluation, energy never reported next to latency, and offline numbers taken from leaky pipelines. Which model to deploy is the other half of the question. We attack all three with one system.

Contributions

The parts are standard (autoencoder, Kafka, Spark MLlib). **No new detector**: the contribution is what we measured on real edge hardware.

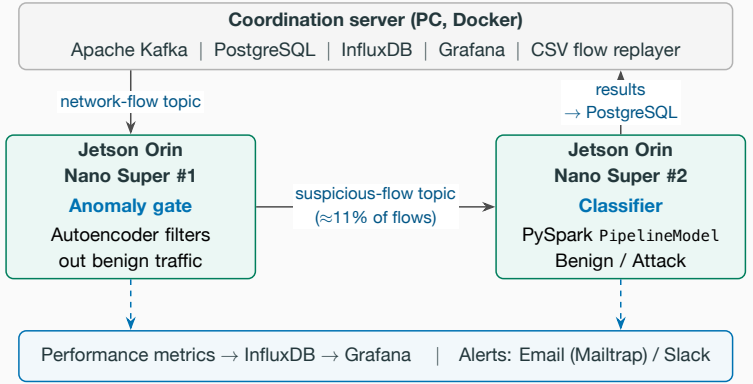
1. A **placement study** of a two-stage IDS on a two-board Jetson cluster: single node, pipeline split on one and on two boards, horizontal Kafka scaling, Spark standalone cluster.
2. An **end-to-end evaluation**: sustained capacity, latency with queueing, energy per verdict, and **precision / recall / F1 after the gate** — on CICIDS2017 and on CSE-CIC-IDS2018 replay.
3. A **same-model engine baseline**: PySpark vs. NumPy vs. ONNX Runtime on the deployed forest.
4. **Leakage-aware provenance**: the deployed model is the one selected by our offline Spark ML evaluation under a leakage-free protocol.


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We do not claim a new detector or a new distributed method. The contribution is the measurement: which placement pays, why, and what the gate costs in detection. The model we deploy is the one that survived a strict offline audit.

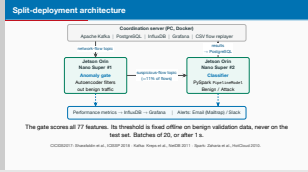
II. SYSTEM AND EVALUATION SETUP

Split-deployment architecture



The gate scores all 77 features. Its threshold is fixed *offline* on benign validation data, never on the test set. Batches of 20, or after 1 s.

CICIDS2017: Sharafaldin et al., ICISSP 2018 · Kafka: Kreps et al., NetDB 2011 · Spark: Zaharia et al., HotCloud 2010.



Host runs the infrastructure. Board one gates, board two classifies. Only suspicious flows cross the Kafka topic between them. Everything is instrumented into Grafana.

Deployment modes and what we measure

Single node full pipeline on one board, gate off or on.

A: Pipeline split gate on #1, classifier on #2 (our design).

A1: Split, one board gate and classifier as two processes on #2.

B: Horizontal both boards run the full role (gate off).

C: Spark cluster host as master, both boards as workers.

Every flow is followed from send to verdict, queueing included. **Sustained capacity**: the highest rate at which every flow gets its verdict within 10 s of the load.

Setup

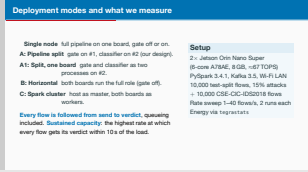
2× Jetson Orin Nano Super (6-core A78AE, 8 GB, ≈67 TOPS)

PySpark 3.4.1, Kafka 3.5, Wi-Fi LAN

10,000 test-split flows, 15% attacks + 10,000 CSE-CIC-IDS2018 flows

Rate sweep 1–40 flows/s, 2 runs each

Energy via `tegrastats`



Latency is measured from the moment a flow is sent to the moment its verdict is written, so time spent waiting in a queue counts. A fresh Kafka consumer group per run, so no run inherits another's backlog.

III. RESULTS AND CONCLUSION

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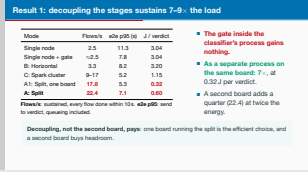
Result 1: decoupling the stages sustains 7–9× the load

Mode	Flows/s	e2e p95 (s)	J / verdict
Single node	2.5	11.3	3.04
Single node + gate	≈2.5	7.8	3.04
B: Horizontal	3.3	8.2	3.20
C: Spark cluster	9–17	5.2	1.15
A1: Split, one board	**17.8**	5.3	**0.32**
A: Split	**22.4**	**7.1**	**0.60**

Flows/s: sustained, every flow done within 10 s. **e2e p95**: send to verdict, queueing included.

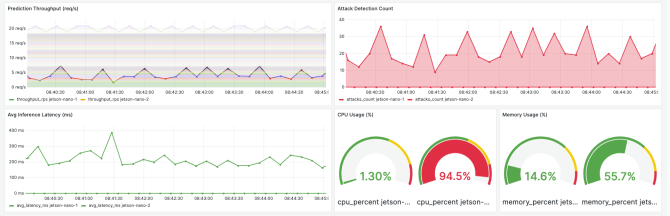
- **The gate inside the classifier's process gains nothing.**
- **As a separate process on the same board: 7×**, at 0.32 J per verdict.
- A second board adds a quarter (22.4) at twice the energy.

Decoupling, not the second board, pays: one board running the split is the efficient choice, and a second board buys headroom.



Single node and horizontal run with the gate off, as in the submitted version. Energy is measured at each mode's sustained rate: 2.0, 2.0, 3.9, 9.5, 17.8 and 18.4 flows per second. Tail latency is seconds in every mode because one PySpark batch takes seconds. A1 was measured in a later session with the same boards, models and replay file. At 30 flows per second one repeat drained in 16 seconds. Gate skip is about 89 percent in every gated mode. Median latency in split mode is about one second. Single node plus gate: the 10 second rule puts it at 1.9 only because one repeat drained in 10.6 seconds; at 3 flows per second it delivers 2.4 to 2.5 like the single node. Spark cluster: at 20 flows per second one repeat kept up at 17.1 and the other did not, so its capacity is only known to lie between about 9 and 17.

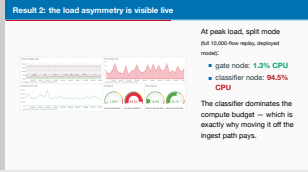
Result 2: the load asymmetry is visible live



At peak load, split mode (full 10,000-flow replay, deployed model):

- gate node: **1.3% CPU**
- classifier node: **94.5% CPU**

The classifier dominates the compute budget — which is exactly why moving it off the ingest path pays.



Grafana at peak load during the full replay. About one versus ninety-five percent. The asymmetry is the whole argument for the split. Add: the gate is a tunable filter, not a fixed point — backup slide has the operating curve.

Result 3: what the gate costs, end to end

Mode	Attacks	Missed: gate / clf	FP	F1
A: full replay	1,516	463 / 0	0	**0.820**
A1: split, one-board	992	279 / 0	0	0.813
C: Spark cluster	381	115 / 0	0	0.822
Gate off	242	– / 0	0	1.000

Final verdicts of the deployed pipeline against the replay labels. Full replay: all 10,000 flows once, 20 flows/s for 500 s. Classifier alone, offline: F1 0.995.

- **Precision 100%**: no false alarm on 8,483 benign flows.
- **Recall 69.5%**: every missed attack was *skipped by the gate*. The classifier made no error on what it received.
- The gate's threshold alone sets the end-to-end F1.

The capacity gain is paid for in recall: the gate is a **tunable trade-off** (operating curve in backup), not a free filter.

The reviewers asked for exactly this: precision, recall and F1 of the whole deployed pipeline, not the offline classifier. The deployed threshold is the 0.995 quantile of benign validation errors, fixed before any test. The capacity-window rows overlap because every window replays the sample from its first row. The full replay covers every flow once.

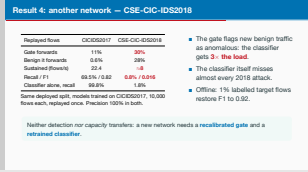
Result 4: another network — CSE-CIC-IDS2018

Replayed flows	CICIDS2017	CSE-CIC-IDS2018
Gate forwards	11%	**30%**
Benign it forwards	0.6%	28%
Sustained (flows/s)	22.4	**≈8**
Recall / F1	69.5% / 0.82	**0.8% / 0.016**
Classifier alone, recall	99.8%	1.8%

Same deployed split, models trained on CICIDS2017, 10,000 flows each, replayed once. Precision 100% in both.

- The gate flags new benign traffic as anomalous: the classifier gets **3× the load**.
- The classifier itself misses almost every 2018 attack.
- Offline: 1% labelled target flows restore F1 to 0.92.

Neither detection *nor capacity* transfers: a new network needs a **recalibrated gate** and a **retrained classifier**.



Reviewer one asked for a second dataset, so we replayed CSE-CIC-IDS2018 through the deployed boards without retraining. Capacity is not a property of the pipeline alone: the gate's skip ratio depends on the traffic, and with it the load on the classifier. At 12 flows per second a repeat drained in 29 seconds. The offline cross-dataset study, 18 runs, gives AUC-PR 0.48 from 2017 to 2018 and 0.19 the other way.

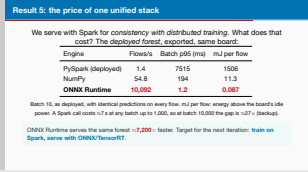
Result 5: the price of one unified stack

We serve with Spark for *consistency with distributed training*. What does that cost? The *deployed forest*, exported, same board:

Engine	Flows/s	Batch p95 (ms)	mJ per flow
PySpark (deployed)	1.4	7515	1506
NumPy	54.8	194	11.3
ONNX Runtime	**10,092**	**1.2**	**0.087**

Batch 10, as deployed, with identical predictions on every flow. mJ per flow: energy above the board's idle power. A Spark call costs ≈7 s at any batch up to 1,000, so at batch 10,000 the gap is ≈27× (backup).

ONNX Runtime serves the same forest ≈**7,200**× faster. Target for the next iteration: **train on Spark, serve with ONNX/TensorRT**.



The JVM costs us more than three orders of magnitude, and that sets the roadmap. Batch size 10. If asked whether bigger batches fix it: a Spark call costs about seven seconds whatever the batch, so the gap shrinks to about 27 times at ten thousand flows, but a streaming IDS at 20 flows per second would wait eight minutes to fill such a batch.

Conclusion

What we showed

- Reproducible edge IDS: Kafka + PySpark on Jetson Orin Nano Super.
- Decoupled stages: **2.5 → 17.8 flows/s** on one board (0.32 J), **22.4** on two (0.60 J), e2e p95 ≈7 s.
- End to end: **precision 100%**, **recall 69.5%**, F1 0.82. Every miss is a gate skip.
- ONNX serves the same forest ≈7,200× faster.

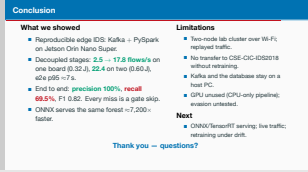
Limitations

- Two-node lab cluster over Wi-Fi; replayed traffic.
- No transfer to CSE-CIC-IDS2018 without retraining.
- Kafka and the database stay on a host PC.
- GPU unused (CPU-only pipeline); evasion untested.

Next

- ONNX/TensorRT serving; live traffic; retraining under drift.

Thank you — questions?


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ONNX/TensorRT serving; live traffic; retraining under drift.
Thank you — questions?

Recap: decoupling gives seven to nine times the load at a fraction of the energy. The price is about thirty percent of attacks skipped by the gate. On another dataset both detection and capacity fall, so a new network needs recalibration and retraining.

Key references

[1] Sharafaldin, I., Lashkari, A. H. & Ghorbani, A. A. *Toward generating a new intrusion detection dataset and intrusion traffic characterization*. ICISSP 2018. **CICIDS2017**

[2] Shi, W. et al. *Edge computing: Vision and challenges*. IEEE Internet of Things Journal 3(5), 2016. **edge setting**

[3] Chandola, V., Banerjee, A. & Kumar, V. *Anomaly detection: A survey*. ACM Computing Surveys 41(3), 2009. **anomaly gate**

[4] Lundberg, S. M. & Lee, S.-I. *A unified approach to interpreting model predictions*. NeurIPS 2017. **SHAP Top-30**

[5] Zaharia, M. et al. *Spark: Cluster computing with working sets*. HotCloud 2010. **Spark**

[6] Kreps, J., Narkhede, N. & Rao, J. *Kafka: a Distributed Messaging System for Log Processing*. NetDB 2011. **Kafka**

[7] Kiran Kumar, K. M. et al. *Distributed Intrusion Detection System using Kafka and Spark Streaming*. ICVADV 2025. **closest system**

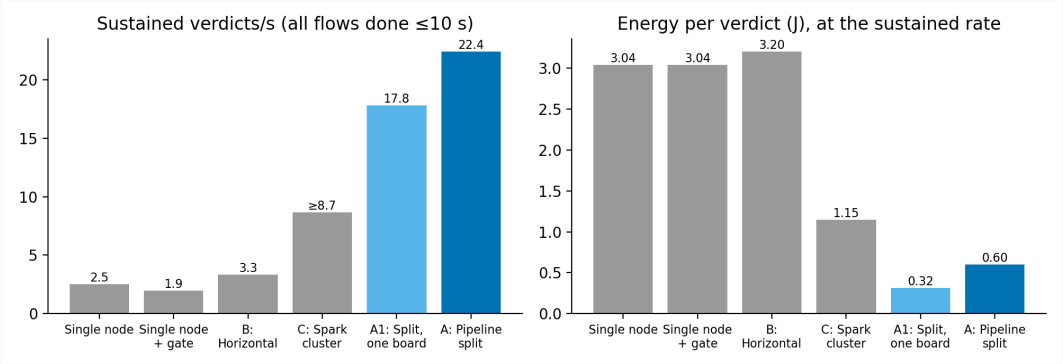
[8] Arp, D. et al. *Dos and Don'ts of Machine Learning in Computer Security*. USENIX Security 2022. **leakage-aware protocol**

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[6] Kreps, J., Narkhede, N. & Rao, J. *Kafka: a Distributed Messaging System for Log Processing*. NetDB 2011. Kafka
[7] Kiran Kumar, K. M. et al. *Distributed Intrusion Detection System using Kafka and Spark Streaming*. ICVADV 2025. closest system
[8] Arp, D. et al. *Dos and Don'ts of Machine Learning in Computer Security*. USENIX Security 2022. leakage-aware protocol

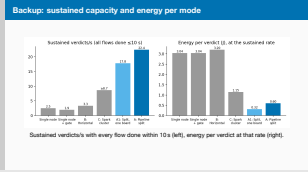
Not presented — shown while taking questions, so sources are on screen if the audience asks. Full list is in the paper.

Backup slides

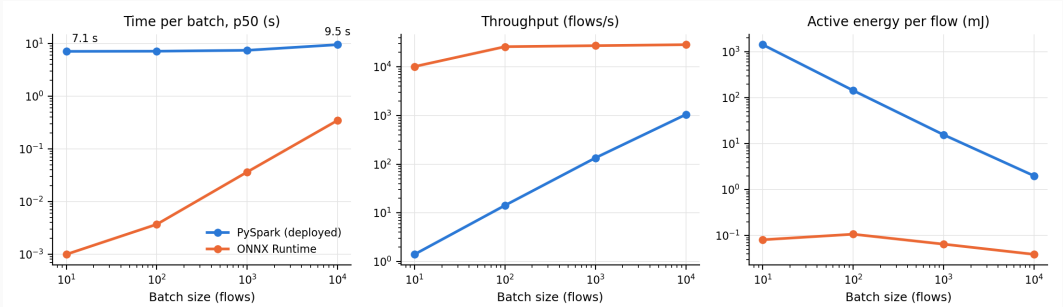
Backup: sustained capacity and energy per mode



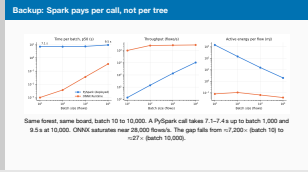
Sustained verdicts/s with every flow done within 10 s (left); energy per verdict at that rate (right).



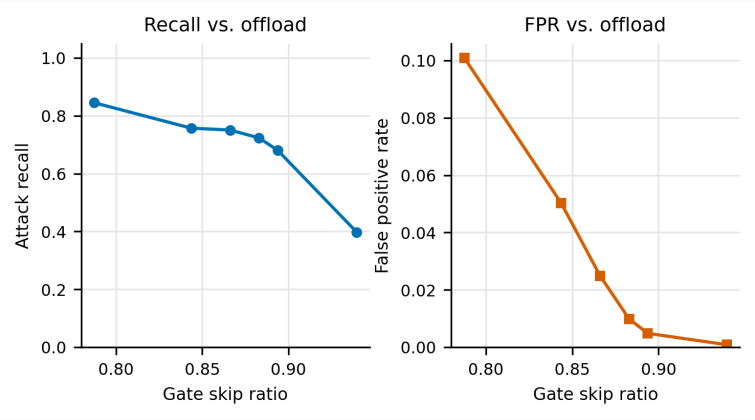
Backup: Spark pays per call, not per tree



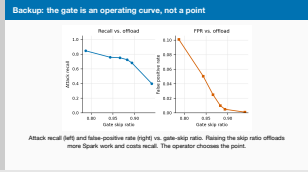
Same forest, same board, batch 10 to 10,000. A PySpark call takes 7.1–7.4 s up to batch 1,000 and 9.5 s at 10,000. ONNX saturates near 28,000 flows/s. The gap falls from ≈7,200× (batch 10) to ≈27× (batch 10,000).



Backup: the gate is an operating curve, not a point



Attack recall (left) and false-positive rate (right) vs. gate-skip ratio. Raising the skip ratio offloads more Spark work and costs recall. The operator chooses the point.



Backup: which model, and why

Feature set	Features	XGBoost F1	RF F1	XGBoost size (MB)	Edge rationale
All features (baseline)	77	**0.9971**	0.9949	2.32	Reference
SHAP Top-30	30	0.9970	**0.9955**	2.37	Explainable; deployed
RF Top-30	30	0.9922	0.9879	2.52	Embedded ranking
PCA k=40	40	0.9939	0.9837	3.82	Unsupervised

We deploy **SHAP Top-30 with Random Forest** (exported model: test F1 0.9959, 30 features): a native Spark ML estimator, so the `PipelineModel` loads on ARM64 without the native libraries Spark's XGBoost integration needs. The SHAP ranking is computed after `destination_port` is removed.

†Measured on a single-node retrain (the cluster save kept only metadata).

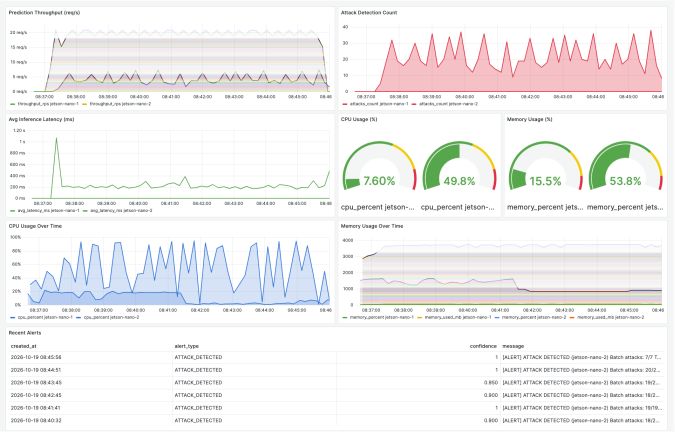
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Backup: full monitoring dashboard



Per-node throughput, detected attacks, inference latency, CPU/RAM for both boards, PostgreSQL-backed alerts.

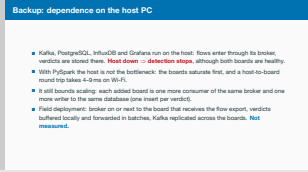
Backup: full monitoring dashboard



Per-node throughput, detected attacks, inference latency, CPU/RAM for both boards, PostgreSQL-backed alerts.

Backup: dependence on the host PC

- Kafka, PostgreSQL, InfluxDB and Grafana run on the host: flows enter through its broker, verdicts are stored there. **Host down ⇒ detection stops**, although both boards are healthy.
- With PySpark the host is *not* the bottleneck: the boards saturate first, and a host-to-board round trip takes 4–9 ms on Wi-Fi.
- It still bounds scaling: each added board is one more consumer of the same broker and one more writer to the same database (one insert per verdict).
- Field deployment: broker on or next to the board that receives the flow export, verdicts buffered locally and forwarded in batches, Kafka replicated across the boards. **Not measured.**



Backup: measurement protocol

- Host-side orchestration starts pipeline roles on the Jetsons over SSH, drives the load, collects every verdict.
- Each verdict stores the flow's send time and replay row: latency is send to verdict, *queueing included*. Skip ratio and attack recall come from the flows themselves.
- Rate sweep per mode. A rate is sustained when every flow sent in the 45 s window gets its verdict within 10 s after it. 2 runs per rate.
- Fresh Kafka consumer group per run, so no run inherits a backlog.
- Tiers batch 20 flows or flush after 1 s.
- Energy: `tegrastats`, 30 s idle baseline then load at the sustained rate. Two-node modes sum both boards.

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